



SYMBIOSIS INSTITUTE OF TECHNOLOGY

(Established under section 3 of the UGC Act 1956) Re-accredited by
NAAC with “A” Grade

NAME – SOURABH GUPTA

PRN :25070122191

Batch – B3

COURSE – JAVA

Experiment 9-

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

AIM – Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Program 9

```
// Program to demonstrate the final keyword in Java

// Final class (cannot be inherited)
final class College {

    void displayCollege() {
        System.out.println("College Name : SIIT Pune");
    }
}

// Parent class
class Animal {

    // Final variable (constant)
    final int MAX_MARKS = 200;

    // Final method (cannot be overridden)
    final void sound() {
        System.out.println("Animal makes different sound");
    }

    void displayMarks() {
        System.out.println("Maximum Marks = " + MAX_MARKS);
    }
}

// Child class
class Dog extends Animal {

    void displayDog() {
        System.out.println("Dog barks");
    }

    /*
    // This is NOT allowed because sound() is final.
    // Uncommenting it will give a compile-time error.

    void sound() {
        System.out.println("Dog barks");
    }
    */
}

// Main class
public class FinalKeywordDemo {
```

```
public static void main(String[] args) {  
  
    Dog d = new Dog();  
  
    d.displayMarks();  
    d.sound();  
    d.displayDog();  
  
    College c = new College();  
    c.displayCollege();  
}  
}
```

Output

```
PS C:\Users\cl404_17\javav\java> java FinalKeyword  
Maximum Marks = 200  
Animal makes different sound  
Dog barks  
College Name : SIIT Pune  
PS C:\Users\cl404_17\javav\java>
```

^ ↺ 🔇 ENG 10:19
IN 14-08-2026

EXERCISE

6.1 . Create a Bank Account program where account number is final and cannot be changed

Program 6.1 -

```
class BankAccount {  
  
    // Final variable  
    final int accountNumber;  
  
    String accountHolder;  
    double balance;  
  
    // Constructor  
    BankAccount(int accountNumber, String accountHolder, double balance) {  
        this.accountNumber = accountNumber;  
        this.accountHolder = accountHolder;  
        this.balance = balance;  
    }  
  
    // Method to display account details  
    void display() {  
        System.out.println("Account Number : " + accountNumber);  
        System.out.println("Account Holder : " + accountHolder);  
        System.out.println("Balance          : Rs. " + balance);  
    }  
  
    public static void main(String[] args) {  
  
        BankAccount account = new BankAccount(  
            123456789,  
            "Sourabh Gupta",  
            25000  
        );  
  
        account.display();  
  
        // account.accountNumber = 987654321;  
        // Error: cannot assign a value to final variable accountNumber
```

```
}  
}
```

Screenshot 6.1 –

```
PS C:\Users\cl404_17\javav\java> javac BankAccount.java  
PS C:\Users\cl404_17\javav\java> java BankAccount  
Account Number : 123456789  
Account Holder : Sourabh Gupta  
Balance        : Rs. 25000.0  
PS C:\Users\cl404_17\javav\java> |  
  
^ ↺ ⏻ ENG 10:26  
IN      14-08-2026
```

Program 6.2- Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```
class LibraryBook {  
  
    // Final variable  
    final String ISBN;  
  
    String title;  
    String author;  
    double price;  
  
    // Constructor
```

```
LibraryBook(String ISBN, String title, String author, double price) {
    this.ISBN = ISBN;
    this.title = title;
    this.author = author;
    this.price = price;
}

// Method to display book details
void display() {
    System.out.println("ISBN    : " + ISBN);
    System.out.println("Title   : " + title);
    System.out.println("Author  : " + author);
    System.out.println("Price  : Rs. " + price);
}

public static void main(String[] args) {

    LibraryBook book = new LibraryBook(
        "978-81-12345-67-8",
        "Java Programming",
        "James Gosling",
        599
    );

    book.display();

    // ISBN cannot be changed because it is final
    // book.ISBN = "978-81-98765-43-2";
}
}
```

Screenshot 6.2 –

```
PS C:\Users\cl404_17\javav\java> javac LibraryBook.j
PS C:\Users\cl404_17\javav\java> java LibraryBook
ISBN    : 978-81-12345-67-8
Title   : Java Programming
Author  : James Gosling
Price   : Rs. 599.0
PS C:\Users\cl404_17\javav\java> |
```



ENG
IN



10:37
14-08-2026